

A decorative graphic on the left side of the slide, consisting of a network of white lines and small circles on a blue gradient background, resembling a circuit board or a stylized tree structure.

CE 3372 WATER SYSTEMS DESIGN

LESSON 22: DUAL-DRAINAGE; PUMPS AND LIFT STATIONS

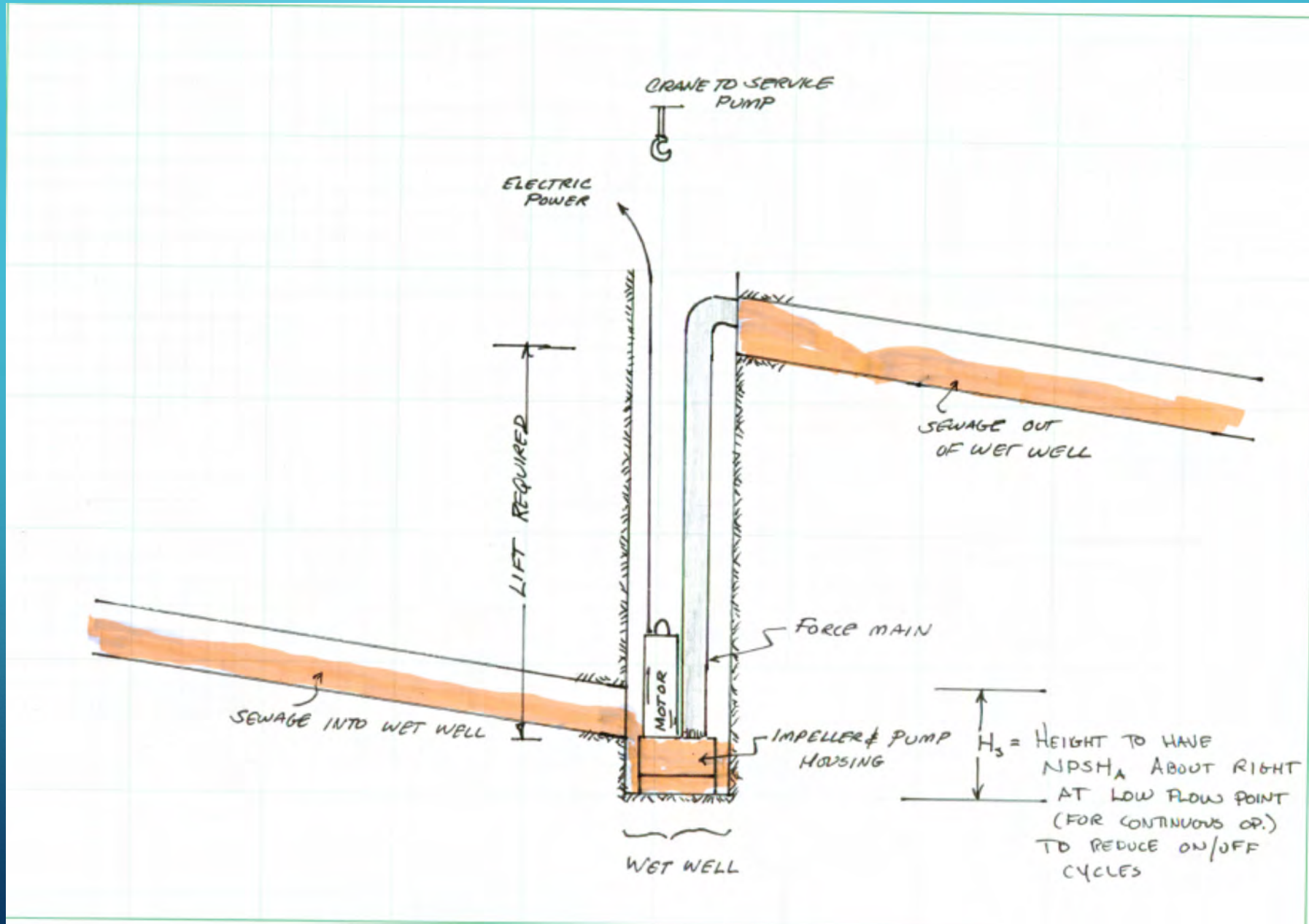
OUTLINE

- Dual-Drainage Systems
 - Simulate Overland and Conduit Flow
 - Use Inlet-Outlet Offsets and multiple connections to each junction
- Lift stations
 - Used to “lift” water from a low elevation to a higher elevation.

OUTLINE

- Lift stations
 - Lift provided by pumps.
 - Pumps modeled similar to EPA-NET
 - PUMPS are LINKS
 - The Sump is often a storage node

CONCEPTUALIZATION



LIFT STATIONS IN SWMM

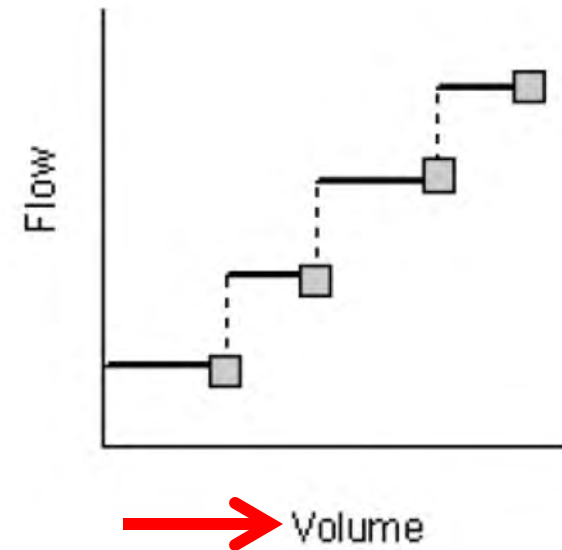
- Lift stations are modeled as special links (like in EPA NET).
- Page 40 of the SWMM manual described the conceptual model used – there are 4 “kinds” of pumps in SWMM

LIFT STATIONS IN SWMM

- Type 1 Pump Rule

Type1

An off-line pump with a wet well where flow increases incrementally with available wet well volume



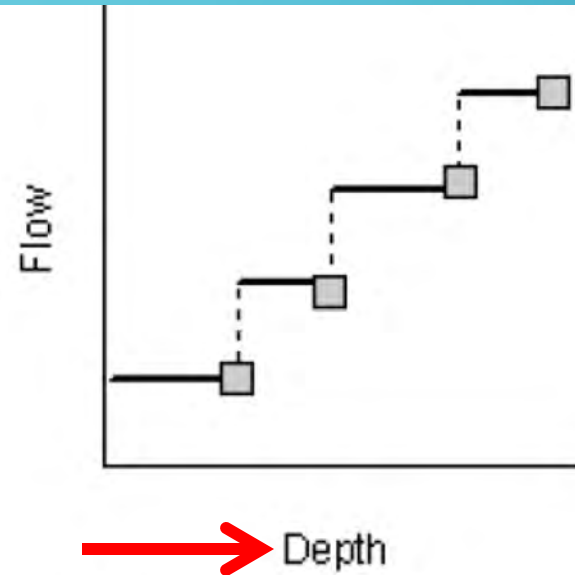
- Semi-realistic (see next slide); actual operation would likely be by stage (depth)

LIFT STATIONS IN SWMM

- Type 2 Pump Rule

Type2

An in-line pump where flow increases incrementally with inlet node depth.



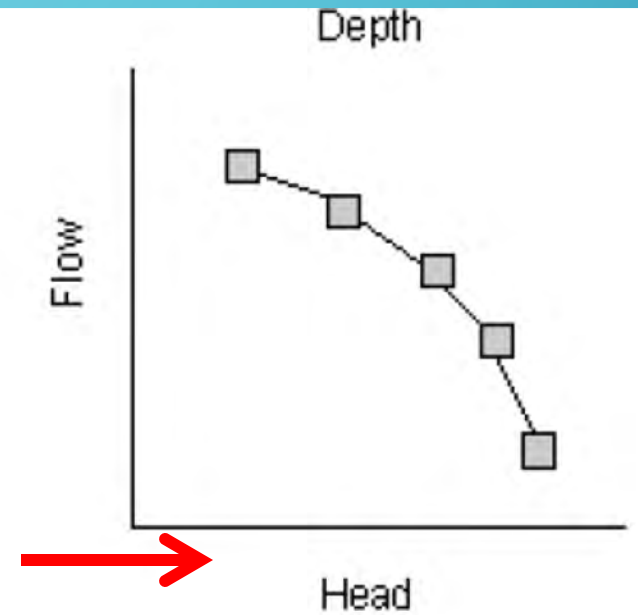
Realistic rule: can use a PLC and transducer to control the pump;
assumes discharge into free (open) pipe

LIFT STATIONS IN SWMM

- Type 3 Pump Rule

Type3

An in-line pump where flow varies continuously with head difference between the inlet and outlet nodes.



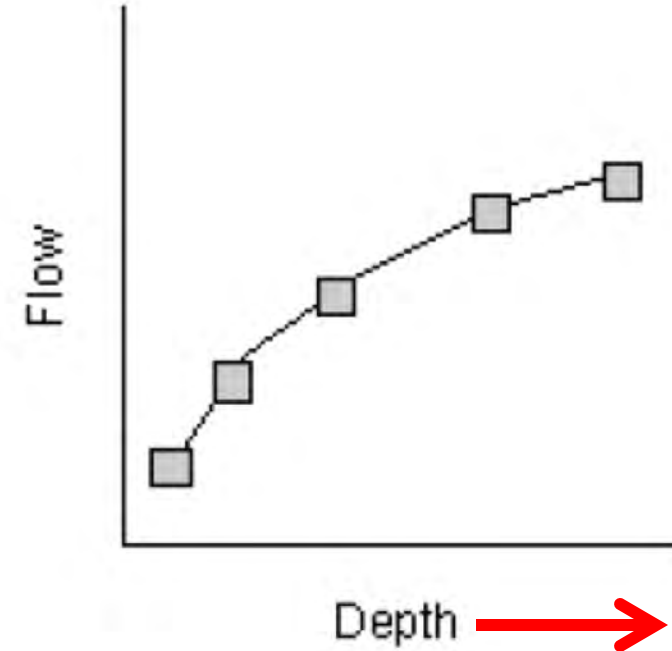
Realistic rule. Like in EPA-SWMM. Assumes suction and discharge submerged – discharging into a force-main. This is the “kind” of pump you had in fluids and earlier in this course.

LIFT STATIONS IN SWMM

- Type 4 Pump Rule

Type4

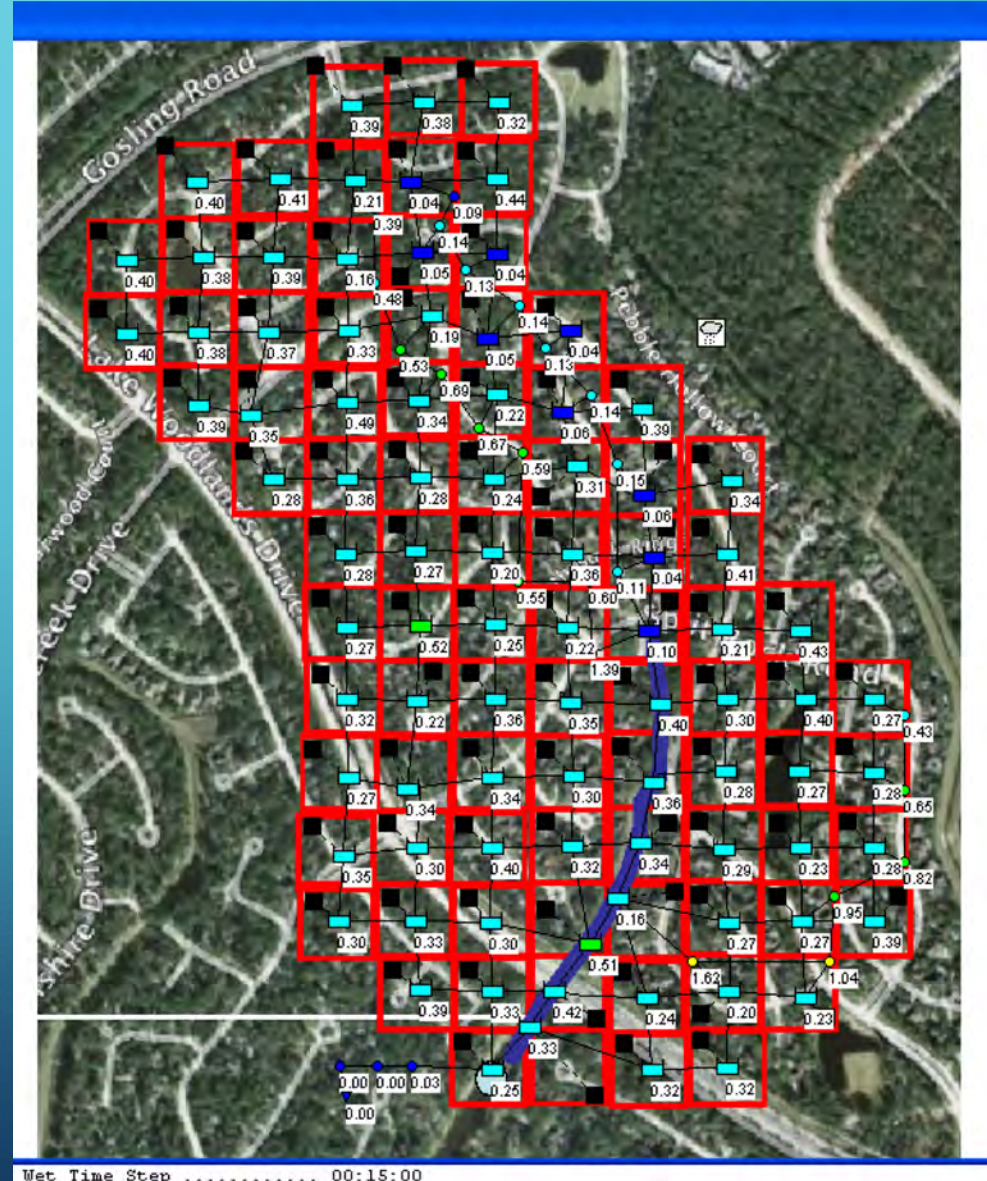
A variable speed in-line pump where flow varies continuously with inlet node depth.



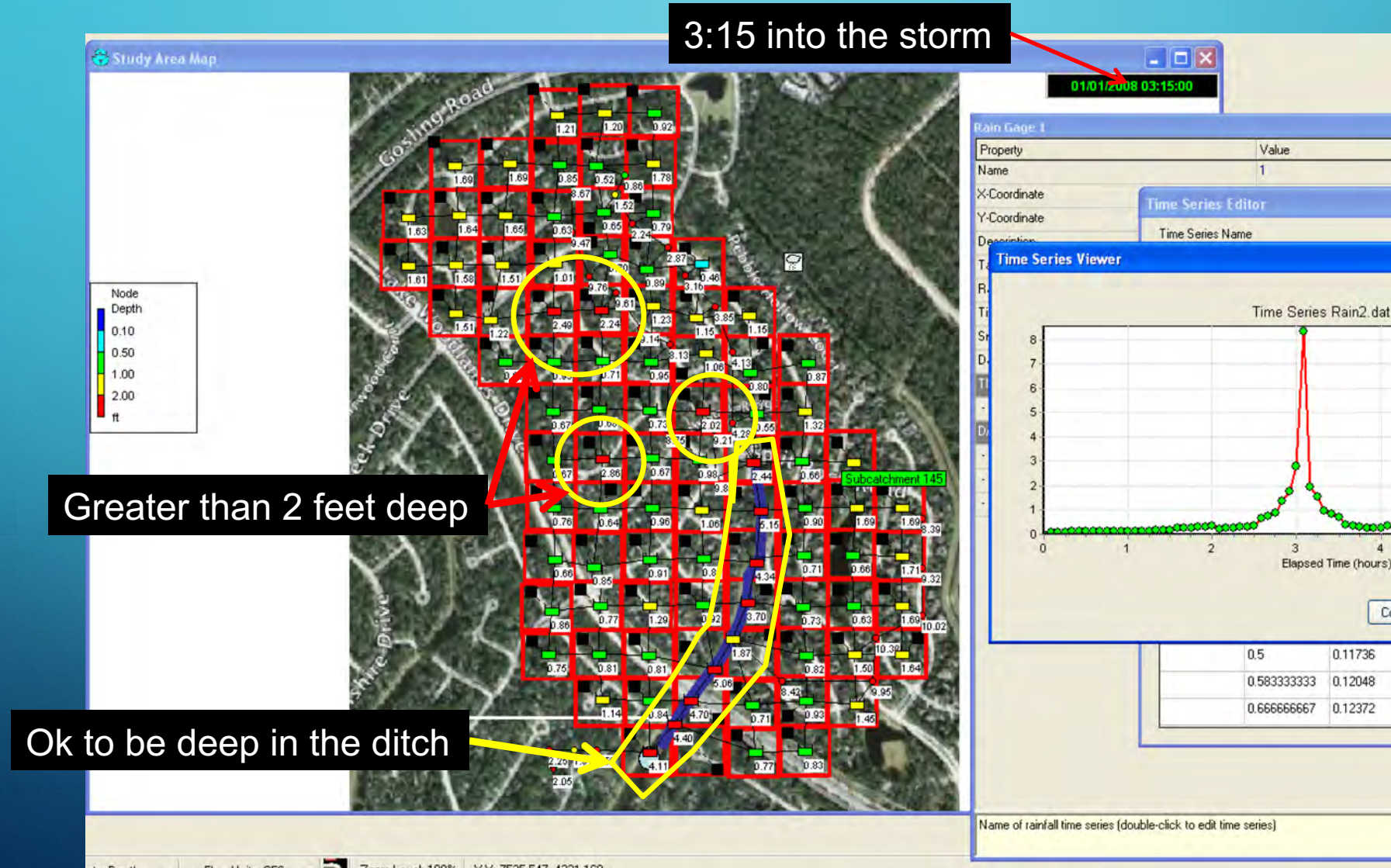
Reasonably common in wastewater treatment plants for initial lift into the reactors (then gravity flow rest of plant).

LIFT STATIONS IN SWMM

- Woodlands Ditch C Subdivision.
 - As-is drainage model, using 2-yr design storm.
 - Suppose the goal is to keep ponding less than 2 feet.

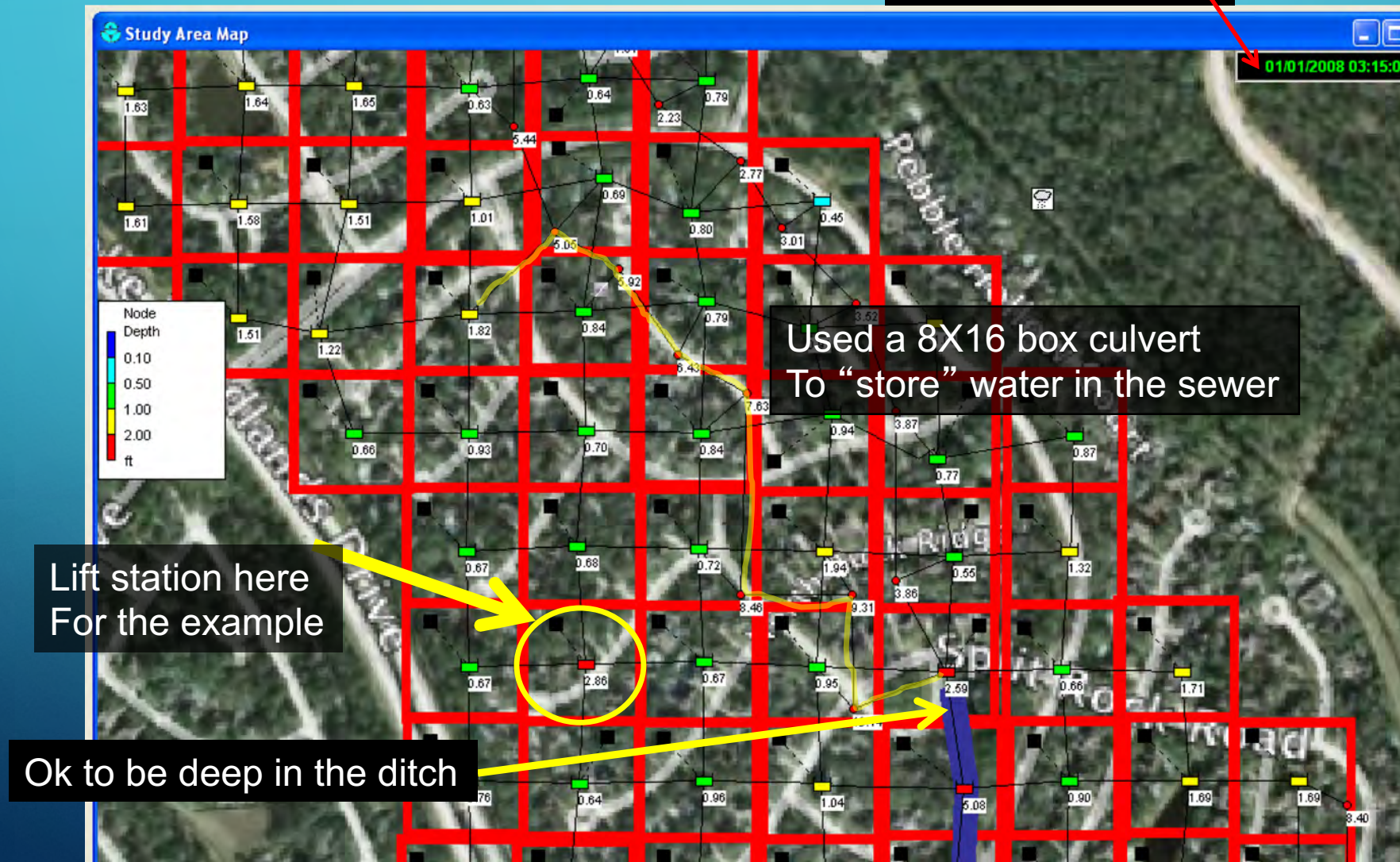


LIFT STATIONS IN SWMM



ADD RELIEF SEWERS

3:15 into the storm

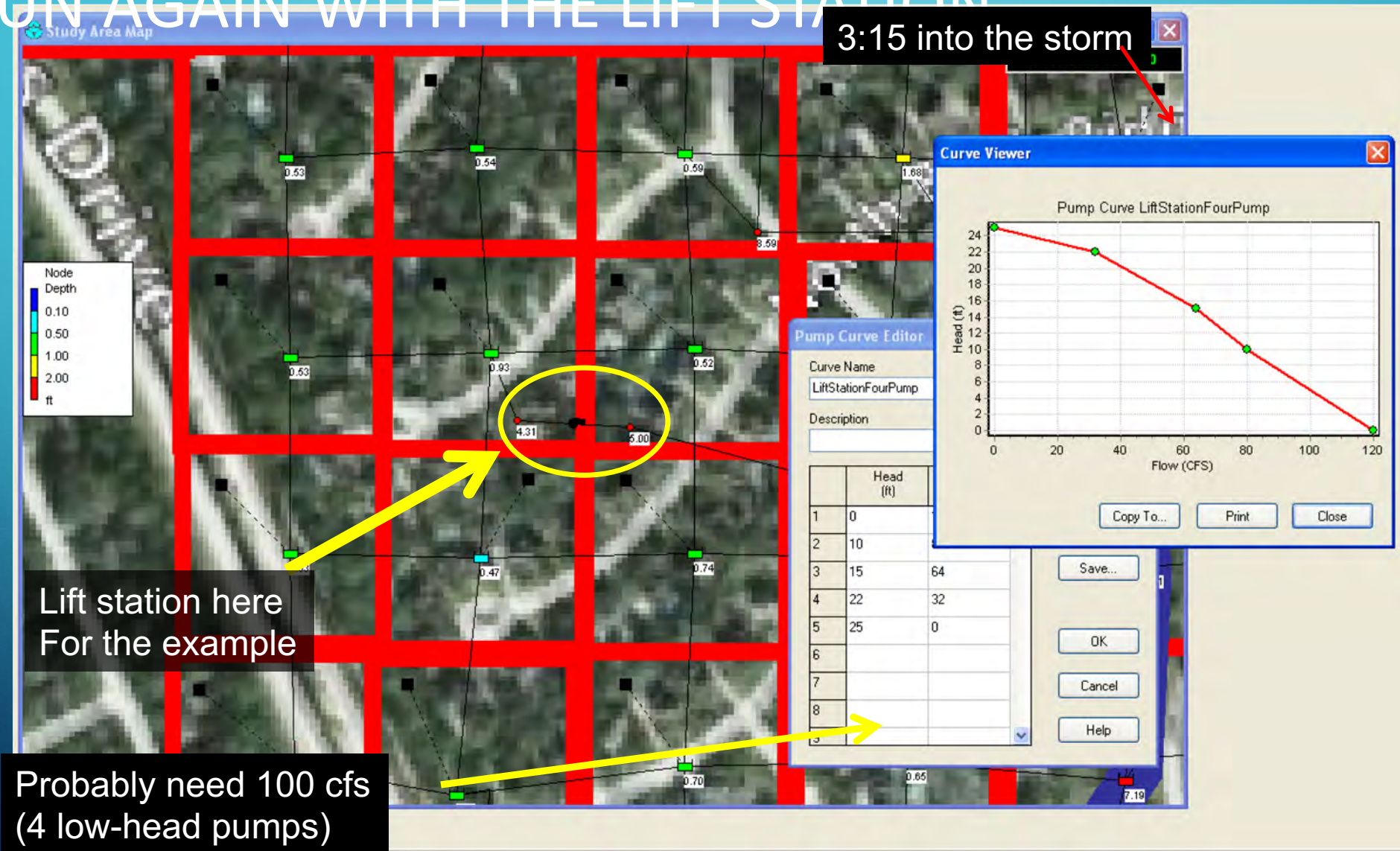


RUN ONCE TO FIND LIFT STATION NEED

3:15 into the storm



RUN AGAIN WITH THE LIFT STATION



THEN CHECK

- Is such a lift station feasible?
 - How many pumps?
 - Force main diameter (4-foot this simulation), kind of big
 - How much power?
 - Other alternatives (in this case a gravity relief sewer probably makes more sense)

JONES STADIUM LIFT STATION IN SWMM

- EXAMPLE OF ENCODING A LIFT STATION INTO SWMM



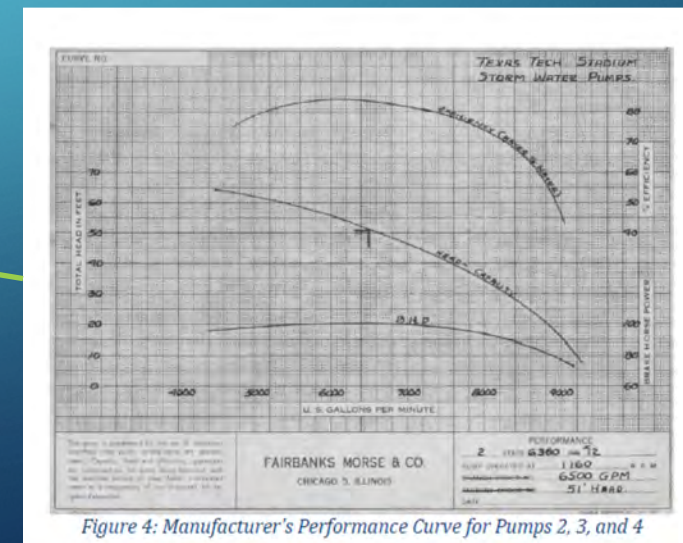
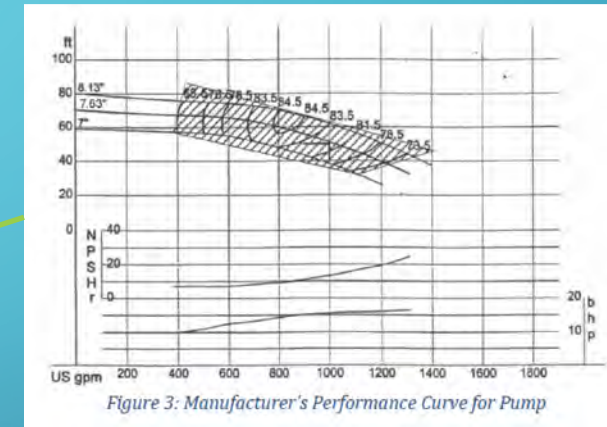
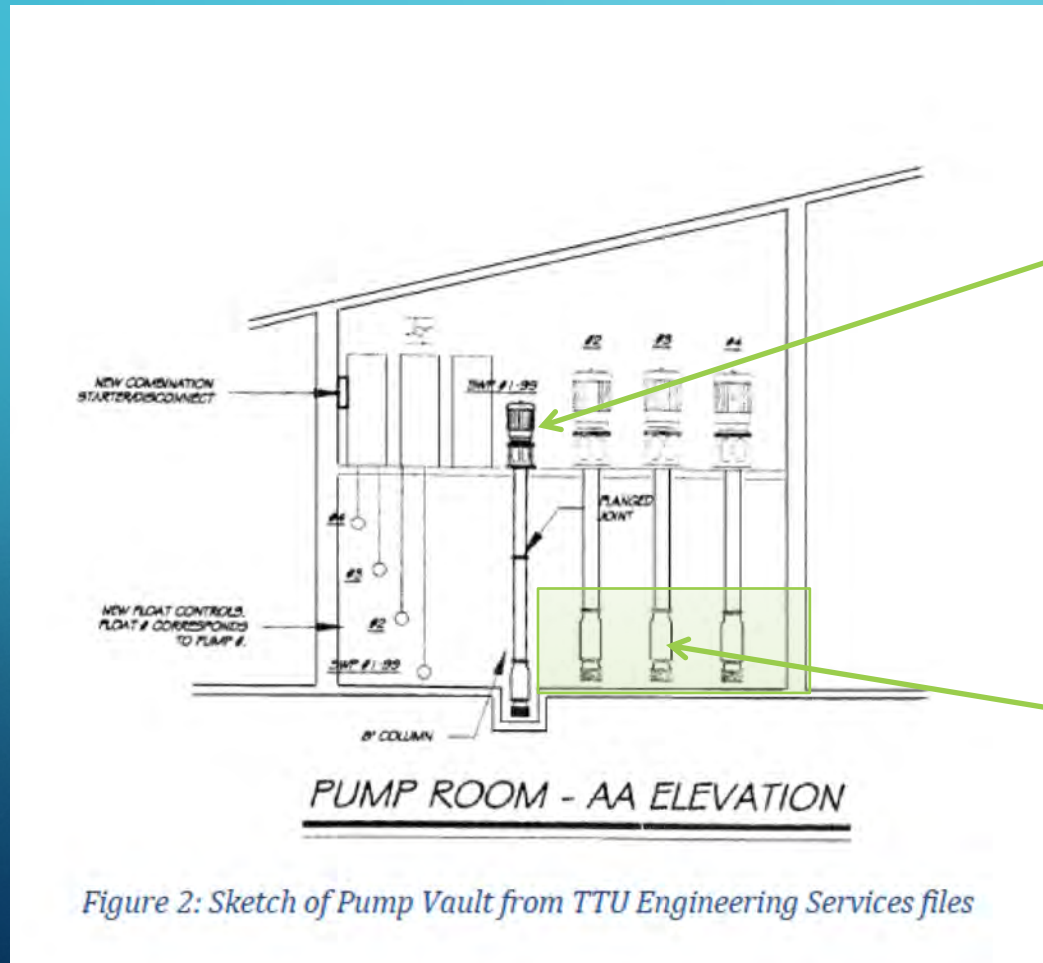
Figure 1. Photographs of Lift Station Location and Field Inlets



Figure 5: Photo Views within the Pump Vault

JONES STADIUM LIFT STATION IN SWMM

- EXAMPLE OF ENCODING A LIFT STATION INTO SWMM



JONES STADIUM LIFT STATION IN SWMM

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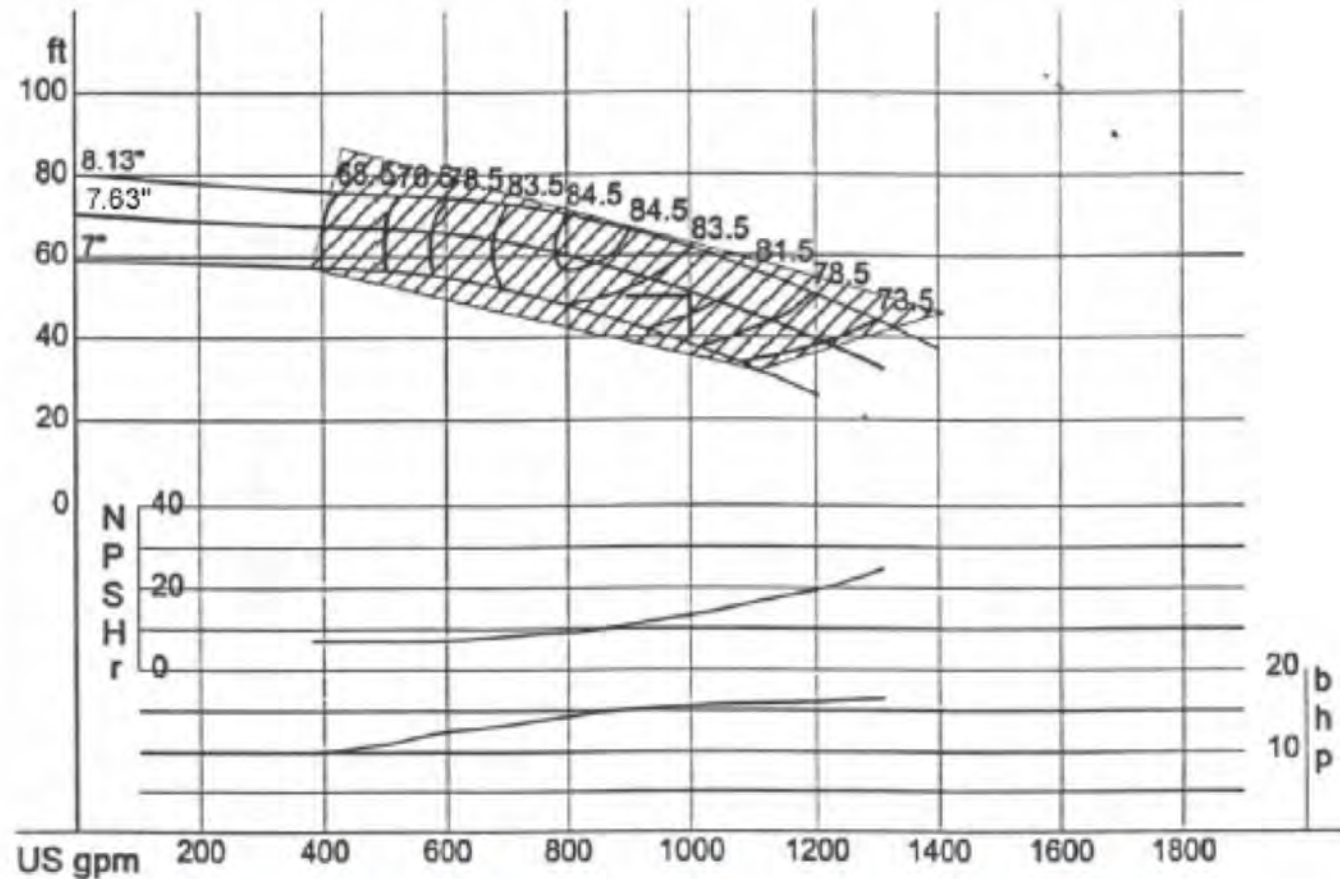


Figure 3: Manufacturer's Performance Curve for Pump

JONES STADIUM LIFT STATION IN SWMM

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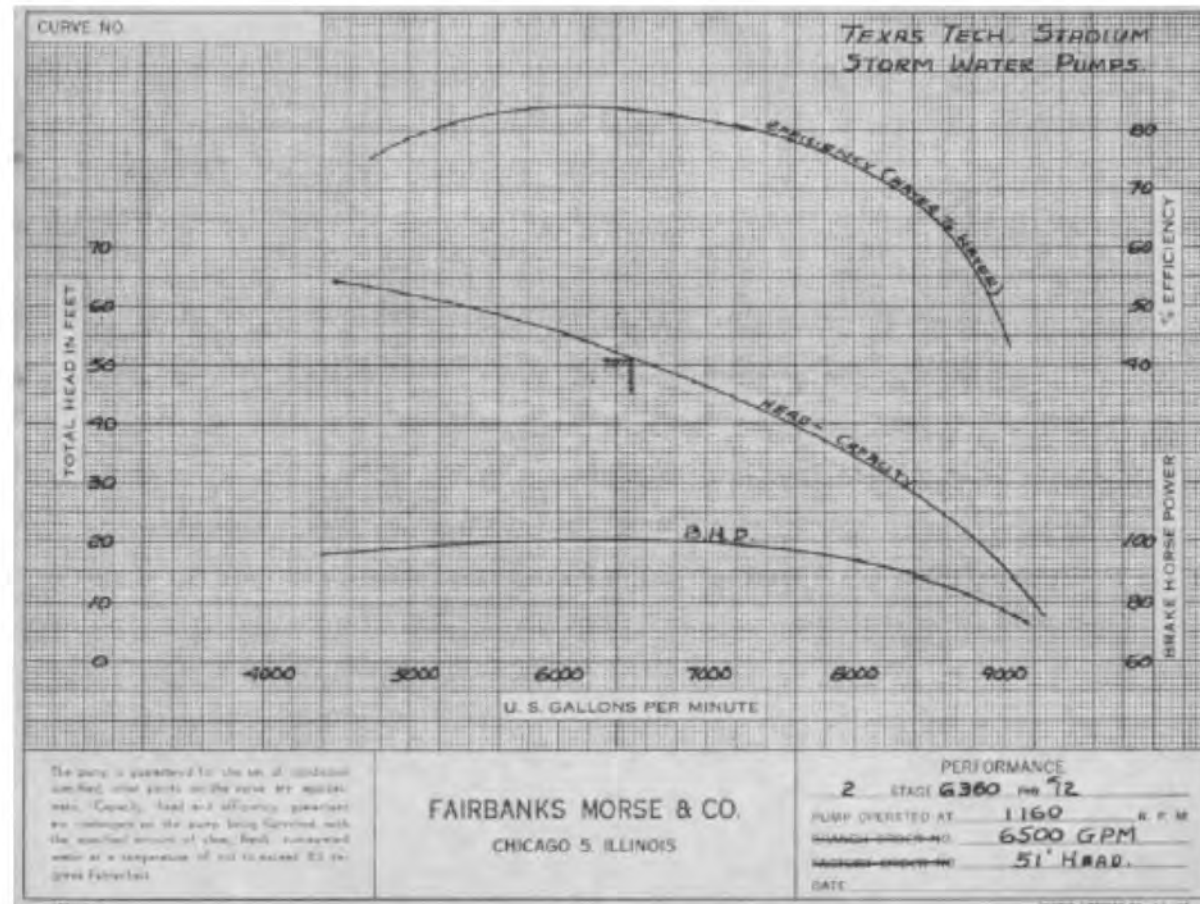
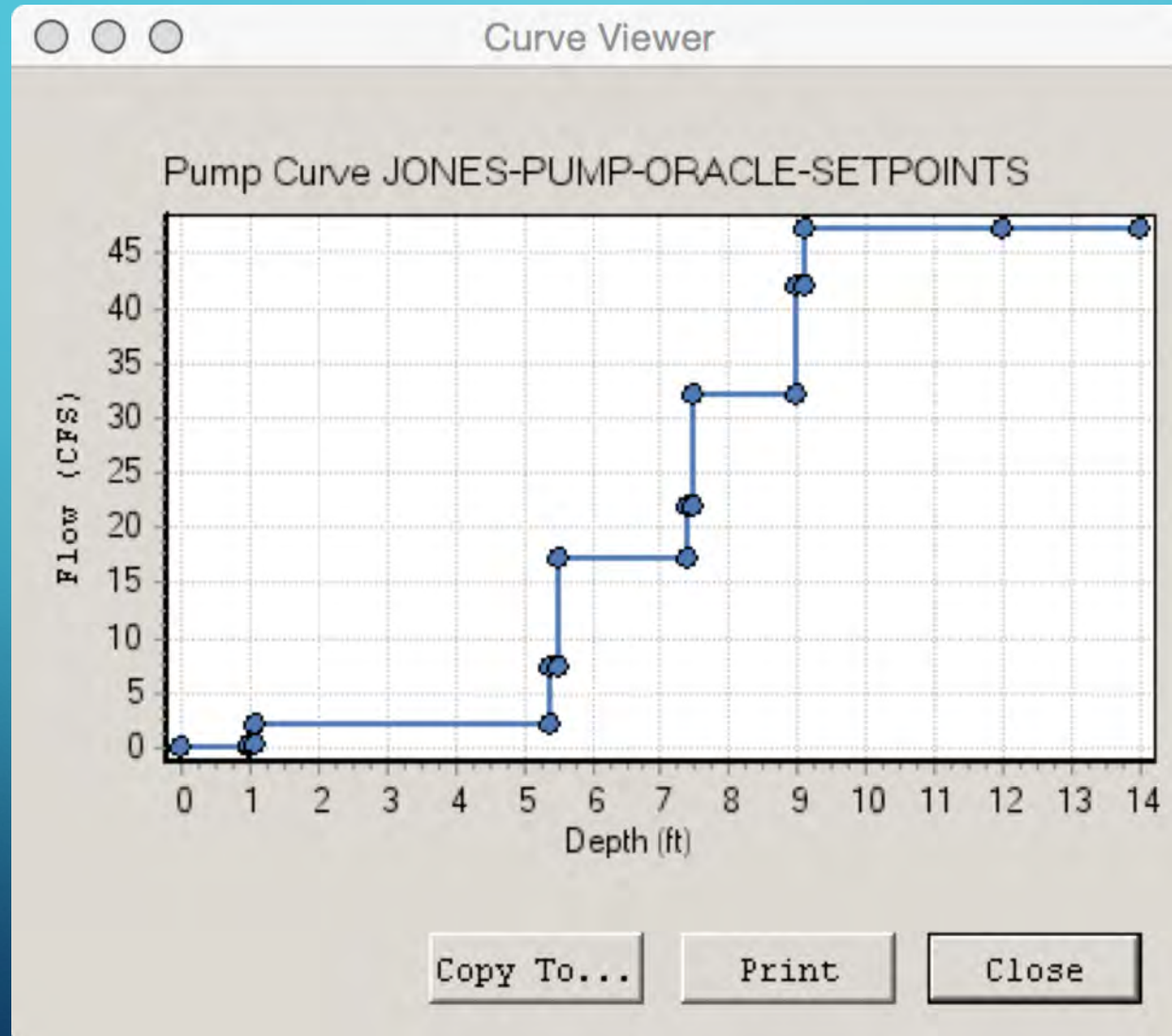


Figure 4: Manufacturer's Performance Curve for Pumps 2, 3, and 4

JONES STADIUM LIFT STATION IN SWMM



JONES STADIUM LIFT STATION IN SWMM

- RUN THE MODEL (TAKES 5-MINUTES) AND CAN EXAMINE THE EFFECT OF THE PUMP(S)